MILITARY
SPECIFICATION

MIL-P-12965(Sig C)
10 August 1953
Superseding U. S. Army
Specification 71-1640
8 April 1946

POWER SUPPLY (RECTIFIER POWER UNIT RA-61-( )) AND MOUNTING FT-414

1. SCOPE

1.1 This specification covers the requirements for one type of rectifier power supply designated as Rectifier Power Unit RA-61( ), and one type of shock mounting for Rectifier Power Unit RA-61-( ) designated as Mounting FT-414. (See 6.1.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND DRAWINGS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591	Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment).
NN-B-601	Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621	Boxes; Wood, Nailed and Lock-Corners.
LLL-B-631	Boxes, Fiber, Corrugated (For Domestic Shipment).
UU-T-111	Tape; Paper, Gummed (Sealing and Securing).

MILITARY

MIL-T-27	Transformers and Inductors (Audio, Power, and Pulse), Plastic-Material, Thermosetting, Cast: For Use in.
JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood Cleated, Plywood.
JAN-P-106	Boxes; Wood, Nailed (Military Type).
MIL-B-107	Boxes, Wood, Wire-Bound, (Overseas Type).
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116	Preservation, Methods Of.
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant.
MIL-B-131	Barrier-Material; Water-Vaporproof, Flexible.
JAN-T-152	Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment; General Process For.
MIL-C-5015	Connectors; Electrical.
MIL-T-10513	Tropicalization of Materials Used in Signal Corps Equipment.
MIL-P-11268	Parts, Materials, and Processes Used in Signal Corps Equipment.

Source: https://dscsit.dla.mil - Downloaded: 2026-10-17T23:54Z
Check the source to verify that this is the current version before use.

U. S. ARMY

19-10	Filler, Interior Packing, Fiberboard, Corrugated, Single- and Double-Faced, for General Packing Purposes.
71-4944	Identification Marking of Ground Signal Equipment.
71-4945	Rubber Substitutes for Use in Cords, Cordages, and Cables.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking of Shipments.

DRAWINGS

SIGNAL CORPS

SC-D-6642	Mounting FT-414
SC-D-15914	Moisture Resistance, Test Cycle for Ground Signal Equipment.
SC-D-17869	Caution Plates.
SC-A-21825	Identification Tag.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions may be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Parts, materials, and processes; general.- In addition to any requirements of this specification, covering parts, material and processes, such items shall conform to Specification MIL-P-11268(Sig0) including the selection requirements therein.

3.2 Procurement model.- The procurement model shall be understood as the one which, in accordance with the invitation for bids, will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein, the equipment shall conform to the following:

3.2.1 Construction.- Construction of the equipment shall conform to the procurement model.

Source: https://assist.dla.mil -- Downloaded: 20261017T23:54Z
Check the source to verify that this is the current version before use.

Parallel operation

3.2.2 Functions.- The equipment shall incorporate all functions provided by the procurement model.

3.2.3 Performance.- Performance of the equipment shall be at least equal to that of the procurement model. When any performance characteristic of the procurement model is not specified herein, the following shall apply: As soon as the procurement model is received by the contractor, he shall adjust it for optimum performance and prepare to conduct all measurements necessary to determine the performance characteristics which are not specified herein. The contracting officer shall be notified of the date when the measurements are to be taken, sufficiently in advance to enable his representative to witness them. Measurements shall be taken by the contractor and three record copies thereof furnished to the contracting officer.

3.3 Preproduction samples.-

3.3.1 Quantities.- After award of contract, the contractor shall furnish two complete Power Supply Units RA-61-( ) and two Mountings FT-414 (if on order), for performance testing.

3.3.2 Compliance with specified requirements.-

3.3.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.3.2.3 for exception.)

3.3.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.3.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.3.2.3 When deviation from 3.3.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and the manner in which it will be corrected in production of equipment on contract.

3.3.3 Test data.- The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.3.4 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of equipments.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:54Z
Check the source to verify that this is the current version before use.

3.3.5 Preproduction inspection.- Preproduction samples may be submitted by the Government to Group A inspection and nondestructive Group B inspection specified in Tables II and III, respectively, and to the preproduction inspection specified in Table I, below. Unlisted nondestructive inspection on preproduction samples of equipment and unlisted destructive or nondestructive inspection on separate preproduction samples of parts or materials may be performed to determine compliance with specified requirements. Preproduction sampling will normally be performed in this order: (1) Vibration, (2) Bounce; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I. - Preproduction inspection

Inspection	Spec. Para.	Test Para.
Vibration	3.20	4.2
Bounce	3.20	4.3
Extreme Temperature	3.9	4.4
Humidity	3.9	4.5

3.4 Circuit label.- A copy of the complete circuit diagram of Power Supply Unit RA-61-( ) shall be mounted on the inside of the hinged door of the case and shall conform to the following:

3.4.1 Lettering.- Lettering shall be in vertical Gothic type, with a minimum height of 3/64 inch.

3.4.2 Border.- Where space permits, a border of 1/4 inch or more shall be provided on each edge of the label.

3.4.3 Material and process.- Material and process for labels shall conform to the following:

(a) Aluminum, brass, or copper, applied by reverse etching, machine engraving, die stamping, lithographing, printing, or silk screening.

(b) Printed on white book paper laminated between two sheets of minimum 0.015-inch, clear, transparent, polyvinyl-chloride-acetate plastic sheets. The two sheets shall be bonded together to seal against moisture.

(c) Printed on a minimum 0.020-inch, white, opaque, polyvinyl-chloride-acetate plastic sheet and covered with a minimum 0.010-inch, clear transparent sheet of the same material. The two sheets shall be bonded together to seal against moisture.

3.4.4 Permanency and legibility.- Labels shall meet the applicable permanency and legibility requirements and associated tests of Specification 71-4944. However, the abrasion test shall apply only to labels mounted where they are subject to abrasion during normal use of the equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:54Z
Check the source to verify that this is the current version before use.

3.4.5 Mounting.- Labels shall be securely and permanently mounted by screws or rivets that will not stain the labels under specified service conditions.

3.5 Technical manual caution notice.- A technical manual caution notice conforming to Drawing SC-D-17269 shall be provided on the equipment.

3.6 Marking of type numbers on cords.- The type number portion of the nomenclature of Cords CD-872 and CD-873, shall be marked on each cord assembly by moulding or by use of an identication tag conforming to Drawing SC-A-21825 and fastened securely around the cable.

3.7 Tropicalization.- The assembled equipment shall be treated in accordance with Specification JAN-T-152. Before being assembled into the equipment, materials and parts shall be treated in accordance with Specification MIL-T-10513.

3.8 Transformers and coils.- Transformers and coils shall conform to Grade I of Specification MIL-T-27, and, where practical, shall be of the types specified by the supplement thereto. Nonstandard cases and mountings, designated by "YY" shall not be used. The "class" of the transformer or inductor shall be such that the maximum operating temperature, as given in the specification, will not be exceeded when the equipment is subjected to specified service conditions for the equipment.

3.9 Service conditions.- The equipment shall meet the following service conditions:

3.9.1 Equipment operating.- The equipment shall meet the requirements of this specification under any probable combination of the following:

(a) Operation.- Operation from either a 115-volt or 230-volt, 50-60 cycle a-c source of power and supplying the necessary voltages for satisfactory operation of Radio Receivers BC-967-( ), BC-969-( ), BC-974-( ) and BC-1269-( ).

(b) Temperature.- Any ambient temperature in the range from -40°F to +125°F.

(c) Relative humidity.- Any relative humidity up to 95 percent.

3.9.2 Equipment non-operating.- The equipment shall meet the requirements of 3.9.1 after subjection to one one or more of the following non-operating conditions:

(a) Temperature.- Continuous exposure for 72 hours to an ambient temperature of -80°F and for periods as long as 4 hours per day to an ambient temperature of +160°F.

(b) Relative humidity.- Relative humidity up to 100 percent, including condensation caused by temperature changes.

(c) Orientation.- Indefinite storage in any orientation.

(d) Vibration and bounce.- (See 3.20)

Source: https://assist.dla.mil -- Downloaded: 20260-17T23:54Z
Check the source to verify that this is the current version before use.

3.10 Input.- Power Supply Unit RA-61-( ) shall be capable of operation from alternating current voltages of 115 volts ± 15 percent and 230 volts ± 15 percent, at a frequency of 50 to 60 cycles per second. Provision shall be made within the power supply unit case for readily changing from 115 volts to 230 volts.

3.11 Output.- The outputs of Power Supply Unit RA-61-( ) with either of the inputs specified in 3.10 above, shall be 250 volts d-c (plus 15 volts, minus 0 volts) at a current of 120 milliamperes, and 12.6 volts a-c (plus 0.2 volt, minus 0.0 volt) at a current of 3.0 amperes at the receiver end of Cord CD-872. The 12.6 volt a-c filament supply shall be obtained from two 6.3 volt windings (tolerance of plus 0.1 volt, minus 0.0 volt), connected in series. Means shall also be provided for connecting the two 6.3-volt windings either in parallel (for use with 6.3-volt tubes) or in series (for use with 12.6-volt tubes). The voltages shall be measured in the manner prescribed in 4.6.

3.12 Ripple.- The peak a-c ripple (hum) voltage shall not exceed 0.03 percent of the d-c output voltage when a current of 120 milliamperes is being drawn from the d-c supply.

3.13 Voltage regulation.- The voltage regulation of Power Supply Unit RA-61-( ) shall be equal to or better than that of the model. The d-c output voltage shall be within the limits of 180 volts d-c and 320 volts d-c when the input voltage at the input terminal board is varied between 95 volts a-c (or 195 volts a-c) and 135 volts a-c (or 270 volts a-c) with a design-center resistive load that will draw 120 milliamperes (250 volts d-c output with 115 volts a-c input) from the d-c supply. The a-c filament supply voltage output shall be within the limits of 10.7 volts a-c to 15.5 volts a-c when the input voltage at the input terminal board is varied between 95 volts a-c (or 190 volts a-c) and 135 volts a-c (or 270 volts a-c) when a current of 3.0 amperes is being drawn from the a-c filament supply, or shall be within the limits of 5.35 volts a-c to 7.75 volts a-c when the input voltage at the input terminal board is varied between 95 volts a-c (or 190 volts a-c) and 135 volts a-c (or 270 volts a-c) when a current of 6.0 amperes is being drawn from the a-c filament supply. The d-c output voltage shall be within the limits of 195 volts d-c at a current drain of 200 milliamperes, and 300 volts d-c at a current drain of 50 milliamperes when a constant a-c voltage input of 115 volts is applied to the input terminals of Power Supply Unit RA-61-( ), and when the filament supply voltage is 12.6 volts a-c with a current drain of 3 amperes. The voltage regulation shall be tested in accordance with 4.8.

3.14 Case.- Power Supply Unit RA-61-( ) shall be inclosed in a case as per model. Louvres permitting adequate ventilation for the unit shall be provided. The holes for the input and output cables shall be such as to suitably protect the cables from abrasion and cutting. Four catches, two on each end, shall be provided near the bottom of the case to permit the case to be securely mounted on and fastened to Mounting FT-414. Two holes, 1/16 inch in diameter, shall be drilled through the projecting part of each catch. A flexible wire shall be passed through these holes and over the top of the catch to prevent the catch from opening when severe shocks are encountered during transportation.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:54Z
Check the source to verify that this is the current version before use.

3.15 Panel.- The chassis shall be mounted on a panel 4 inches high and 19 inches long to permit the unit to be mounted in a standard 19-inch relay rack. Suitable handles shall be provided to facilitate removal of the chassis from the case.

3.16 Cords.- The cords referenced in 3.16.1 and 3.16.2 shall be furnished with each Power Supply Unit RA-61-( ). They shall be connected to the unit by means of screw-type terminal connections located on insulated strips at the rear of the chassis. The cords shall be impossible to be detached without removing the case cover. The cordage used in these cords shall be insulated with Buna-S compound meeting the requirements of Specification 71-4945.

3.16.1 Cord CD-872.- Cord CD-872 shall carry the power output to a radio receiver. This cord shall be a 7-foot, 6-conductor, shielded power cord, terminating in spade terminals at one end and a type AN-3108-22-55 connector at the other end. The connector shall comply with the requirements of Specification MIL-C-5015. The two conductors to be used for the filament supply shall be 5,530 circular mils (#2 AWG). The other four conductors shall be 1,620 circular mils (#18 AWG). Two of these four conductors shall be for the plate-supply voltage and the remaining two conductors shall be for the power switch circuit (the power switch being in the receiver). Connection between the receiver chassis and power supply chassis shall be made through the shield of the cord. The negative side of the high-voltage supply shall not be grounded to the power supply chassis.

3.16.2 Cord CD-873.- Cord CD-873 shall be used as the input power cord to the power supply unit. This cord shall be a 10-foot, 2-conductor, #16 AWG (2,580 circular mils) power cord terminating in spade terminals at one end and a standard male plug at the other end.

3.17 Electron tube.- The power supply unit shall employ one electron tube, Type JAN-5U4G.

3.18 Fuses.- A 3-ampere and a 1-1/2 ampere fuse shall be provided for the protection of the 115-volt and the 230 volt circuits, respectively.

3.19 Mounting FT-414.- Mounting FT-414 shall be manufactured in accordance with Drawing SC-D-6642.

3.20 Vibration and bounce.-

3.20.1 Vibration.- The equipment shall have no mechanical resonance below 55 cycles per second when shock mounts (if any) are blocked and removed.

3.20.2 Bounce.- The equipment shall be capable of full specification performance and shall not incur any physical damage except surface abrasion after being subjected to the test of 4.3.

3.21 Running spare parts.- Running spare parts shall be furnished as specified by the contract and shall be identical to corresponding parts in the equipment furnished on the order.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:54Z
Check the source to verify that this is the current version before use.

3.22 Bounce conditioning.- After final assembly, each equipment on contract, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by the LAB Corporation, Summit, New Jersey, or equal. The Package Tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this conditioning for one minute. After conditioning the equipment shall be capable of meeting the inspection of Section 4.

3.23 Facsimiles.- In addition to any facsimiles specified by applicable documents, facsimile copies as follows shall be submitted for approval to the contracting officer:

(a) Circuit label.....4 copies required

(b) Caution notice.....2 copies required

3.24 Workmanship.- The equipment, subassemblies, and parts shall be manufactured and assembled in a thoroughly workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection, general.- Government inspection shall consist of Group A and Group B inspection, as specified in 4.1.1 through 4.1.2.2. (See 6.1). Group B inspection shall normally be performed on lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.1.1 Group A inspection.- Group A inspection shall be in accordance with Table II and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

Table II - Government Group A inspection

Inspection or test	Rqt Par.	Test Par.	AQL
Power supply unit			
Input & output voltage	3.11	4.6	1.5 Dphu*
Visual & mechanical, major	---	4.9	1.5 Dphu*
Visual & mechanical, minor	---	4.9	6.5 Dphu*
Mounting FT-414			
Visual & Mechanical, major	---	4.9	1%
Visual & Mechanical, minor	---	4.9	4%

*Defective per hundred units

Source: https://assist.dla.mil -- Downloaded: 20260-17T23:54Z
Check the source to verify that this is the current version before use.

4.1.2 Group B inspection.- Group B inspection shall be in accordance with Table IV.

Table III. - Government Group B inspection

Inspection or Test	Rqt. Par.	Test Par.	Group B Plans
Ripple Voltage	3.12	4.7)	Table IV
Voltage	3.13	4.8)	

4.1.2.1 Order of inspection within Group B.- Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.2.2 Sampling procedure.- Samples shall be selected in an unbiased manner by the Government inspector. Unless otherwise specified, inspection shall be started using Level II of the specified drawing. If the number of defective sample units from a lot exceeds the acceptance number specified on the drawing, the lot shall be rejected. Disposition of the rejected product (sample unit and lots) shall be in accordance with Standard MIL-STD-105 and 4.1.4.

4.1.2.3. Procedure in case of failure.- When a lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and proposed corrective action. If the Government inspector and the contractor cannot agree on the corrective action to be taken, the matter shall be referred to the contracting officer for resolution.

4.1.3 Reinspection of conforming group B sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B inspection, may be accepted on contract, provided that all visible damage incurred during group B inspection is repaired and the units are then subject to and pass group A inspection.

4.1.4 Disposition of nonconforming product.- When defective sample units or rejected lots are resubmitted for acceptance, such items shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the items are resubmitted and shall become the property of the Government.

4.2 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.20, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled to within 10 percent of the specified amplitude. Shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration.

(b) The equipment shall be vibrated successively for two hours in each of three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in one-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the completed structure, of subassemblies and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:54Z
Check the source to verify that this is the current version before use.

Table IV.- Group B-2 limits of the process average

N	Limit A %	Limit B %	Limit C %	N	Limit A %	Limit B %	Limit C T
10	18.78	11.89	*	130-144	8.62	6.91	1.18
11	18.14	11.57	*	145-159	8.62	6.81	1.38
12	17.58	11.29	*	160-179	8.45	6.72	1.55
13	17.09	11.04	*	180-199	8.25	6.62	1.75
14	16.65	10.82	*	200-249	8.08	6.54	1.92
15-16	16.25	10.63	*	250-299	7.76	6.38	2.24
17-18	15.57	10.29	*	300-349	7.52	6.26	2.48
19-20	15.00	10.00	*	350-399	7.33	6.16	2.67
21-23	14.51	9.76	*	400-449	7.18	6.09	2.82
24-26	13.90	9.45	*	450-499	7.05	6.03	2.95
27-29	13.39	9.19	*	500-549	6.95	5.97	3.05
30-34	12.96	8.98	*	550-599	6.86	5.93	3.14
35-39	12.37	8.68	*	600-649	6.78	5.90	3.22
40-44	11.89	8.45	*	650-699	6.71	5.85	3.29
45-49	11.50	8.25	*	700-749	6.65	5.82	3.35
50-59	11.16	8.08	*	750-799	6.59	5.80	3.41
60-69	10.63	7.81	*	800-849	6.54	5.77	3.46
70-79	10.21	7.60	*	850-899	6.50	5.75	3.50
80-89	9.87	7.44	0.13	900-949	6.45	5.73	3.55
90-99	9.59	7.30	0.41	950-999	6.41	5.71	3.59
100-114	9.36	7.18	0.64	1000	6.38	5.69	3.62
115-129	9.06	7.03	0.94				

*Number of sample units included in estimated process average is insufficient for reduced inspection.

N = Number of sample units in estimated process average.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:54Z
Check the source to verify that this is the current version before use.

Table IV .- Group B-2 sampling plans

Lot size /&	Reduced /# Inspection		Normal /# Inspection		Tightened /# Inspection	
	Sample size	Acceptance number	Sample size	Acceptance number	Sample size	Acceptance number
9 to 25, incl.	2	0	2	0	4	0
26 to 65, incl.	2	0	3	0	6	0
66 to 180, incl.	7	1	7	1	17	1
181 to 500, incl	7	1	17	2	28	2
501 to 1,300, incl.	7	1	28	3	41	3
1,301 to 3,200, incl	17	2	41	4	53	4
3,201 to 8,000, incl	17	2	53	5	66	5
8,001 to 22,000, incl.	28	3	66	6	81	6
22,001 and over	28	3	81	7	81	6

/& For lot sizes under 9, the sample size shall be at the option of the Government but not more than 2, and the acceptance number shall be 0.

/# Normal, tightened, or reduced inspection shall be used in accordance with the following: (Part "b" of this table lists values for limits A, B, and C, referenced below.)

(1) Calculate the estimated process average from the formula 100 P/N; where P equals the number of defective units found in N sample units, and N equals the number of sample units included in the estimated process average. The Government will determine the number of consecutive sample units to be used to estimate the process average.

(2) Except as stated in (3) below, use normal inspection as long as the process average is between limits A and C.

(3) Change to tightened inspection whenever the process average becomes equal to or greater than limit A, or whenever a lot fails to pass.

(4) Return to normal inspection from tightened inspection when two lots in succession pass tightened inspection, and the process average becomes less than limit B.

(5) Change to reduced inspection whenever the process average becomes less than limit C.

(6) Return to normal inspection from reduced inspection when the process average becomes equal to or greater than limit C.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T23:54Z
Check the source to verify that this is the current version before use.

4.3 Bounce test.- The equipment in its transit case, shall be placed on the table of the Package Tester as made by the LAB Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The Package Tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. During each 1/2-hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.4 Extreme temperature tests.-

4.4.1 Low temperature tests.- The power supply shall be placed in a cold chamber, at an ambient temperature of -80°F, for a period of at least 72 hours. The temperature shall then be raised to -40°F and the equipment allowed to reach thermal equilibrium. After exposure to the -40°F ambient temperature for at least 24 hours, the equipment shall be operated at this ambient temperature and shall be capable of meeting the performance requirements of this specification.

4.4.2 High temperature test.- The power supply unit shall be placed in a heat chamber, at an ambient temperature of +160°F for a period of at least 4 hours. The temperature shall then be decreased to +125°F and the equipment allowed to reach thermal equilibrium. After exposure to the +125°F ambient temperature for at least 24 hours, the equipment shall be operated at this ambient temperature and shall be capable of meeting the performance requirements of this specification.

4.5 Moisture-resistance test.-

4.5.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements and complete measurements as rapidly as possible.

(c) Test the power supply unit with the case cover closed except during the last two cycles. Perform last two cycles with cover opened.

4.5.2 Test procedure.-

(a) Dry at 130° ± 5°F for 24 hours (with cover open).

(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours with cover open.

(c) Take measurements as specified in 4.5.3 and readjust or realine as necessary to meet full specification requirement.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humdity, and period of time for each portion of the cycle shall conform to Drawing SC-D-15914. The measurements specified in 4.5.3 shall be made during the periods shown on the drawing and shall comply with 4.5.3 (see 4.5.1(c)).

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity. No replacement or repair of parts shall be made. The equipment shall meet the requirements of 4.5.3.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:54Z
Check the source to verify that this is the current version before use.

4.5.3 Performance.- The power supply unit shall be tested for compliance with the requirements specified in 3.11 and 3.13.

4.5.4 Failure.- If any rectifier fails to meet the requirements of 4.5.3 during or after cycling, it does not pass the test.

4.6 Input and output voltage test.- Power Supply Unit RA-61-( ) shall be operated from an a-c supply of 115 volts, 50 to 60 cycles per second. The input voltage and current shall be measured at the male plug of power input Cord CD-873, and the output voltage and currents shall be measured at the receiver end of Cord CD-872. The plate supply voltage output shall be measured with a resistive load on the power supply unit that will draw 120 milliamperes from the d-c supply. The filament supply voltage output shall be measured with a resistive load that will draw 3.0 amperes from the 12-volt filament supply. This test shall be repeated with an a-c voltage input of 230 volts, 50 to 60 cycles per second. The output voltages shall meet the requirements of 3.18.

4.7 Ripple test.- Power Supply Unit RA-61-( ) shall be connected to 115 volt a-c, 50 to 60 cycle per second power source. To measure the percent a-c ripple in the d-c output, the plate supply voltage shall be connected to a resistive load which will draw 120 milliamperes from the power supply. The plate supply voltage shall be connected to the vertical amplifier terminals of an oscilloscope whose amplifier has been calibrated for voltage amplification. The trace on the oscilloscope shall be measured from its midpoint, and the voltage necessary to cause this deflection shall be calculated. The percent voltage ripple can be calculated in terms of the voltage measured on the oscilloscope and the rated d-c output voltage of the power supply unit. The percent of ripple voltage present shall comply with the requirements of 3.12.

4.8 Voltage regulation test.- Connect the input of the power supply unit to an a-c supply with the voltage capable of being varied from 95 to 135 volts. Connect a resistive load in the plate supply output which will draw 50 to 200 milliamperes current, and a resistive load in the filament supply which will draw 3.0 amperes at 12.6 volts a-c or 6.0 amperes at 6.3 volts a-c.

4.8.1 With a constant voltage of 115 volts at the input terminals of Power Supply Unit RA-61-( ), the plate load resistance shall be varied so that a current of 50 to 200 milliamperes is drawn from the power supply. Measurements of plate supply output voltage and current and measurements of filament supply voltage and current shall be made at the receiver end of Cord CD-872. The regulation of the d-c output shall be measured with a filament current of 3.0 amperes and 12.6 volts at all times. The measurements shall comply with the requirements of 3.13.

4.8.2 With a voltage of 115 volts at the input terminals of the power supply unit, connect a resistive load in the plate output supply which will draw 120 milliamperes at the plate supply voltage of 250 volts, and connect a resistive load in the filament supply of 12.6 volts which will draw 3.0 amperes. Vary the a-c input voltage from 95 to 135 volts a-c, keeping the resistive loads at the same value as set above for the design center conditions. Measurement of the input voltage and current shall be made at the input terminal at the rear of power supply unit. Measurement of the plate supply voltage and current and measurement of the filament supply voltage and current shall be made at the receiver end of Cord CD-872. These measurements shall comply with the requirements of 3.13.

Source: https://assist.dla.mil -- Downloaded: 20260-17T23:54Z
Check the source to verify that this is the current version before use.

4.9 Visual and mechanical inspection.- Parts and equipment shall be inspected for workmanship, mechanical fit, loose nuts and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation, Wiring soldered connections, ground connections, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual and mechanical defects similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government inspector as major or minor in accordance with the definitions of Standard MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment.- Power Supply Unit RA-61-( ) and Mounting FT-414 shall be packaged in accordance with Method III of Specification MIL-P-116.

5.1.2 For export shipment.- Preserve and package Power Supply Unit RA-61-( ) and Mounting FT-414 in accordance with procedures specified for the designated methods as prescribed in Specification MIL-P-116, and further described in the following paragraphs.

5.1.2.1 Tubes.- Spare tube shall be mounted in dummy socket on chassis. No packaging required.

5.1.2.2 Fuses.- Place spare fuses within dummy mountings provided on chassis of power supply unit. No packaging required.

5.1.2.3 Power Supply Unit RA-61-( ) and Mounting FT-414.- Package each power unit Method 11b, as follows: Place 1 power unit on mounting, secure fastenings. Cushion each power supply unit on all surfaces with cells or pads or obht fabricated of corrugated fiberboard, conforming to Specification 19-10 and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned power supply unit together with the required amount of desiccant as prescribed in Specification MIL-P-116, within a close-fitting suitable style and grade corrugated fiberboard box. Seal entire closure with gummed Kraft tape conforming to Specification UU-T-111. Blunt all corners of the box. Place the boxed power unit within a water-vapor-proof barrier, qualified for Class 2 application in accordance with the requirements of Specification MIL-B-131. Heat seal closure. Place the water-vapor-proofed power unit within a close-fitting suitable style and grade fiberboard box conforming to Specification JAN-P-108. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape, conforming to Specification JAN-P-127.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:54Z
Check the source to verify that this is the current version before use.

5.2 Packing.-

5.2.1 For domestic shipment.- Power Unit RA-61-( ) shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, or corrugated fiberboard boxes conforming to Specification NN-B-591, NN-B-601, NN-B-621, and LLL-B-631, respectively. Closures shall be made in accordance with the applicable box specification.

5.2.2 For export shipment.- Place a quantity of power unit packaged as specified in 5.1.2.3 together within a wood cleated plywood, nailed wood or wire bound box conforming to the requirements of specifications JAN-P-105, JAN-P-106 and MIL-B-107, respectively. Fabricate the boxes to fit contents snugly. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping container as specified in the appendix of the applicable box specification.

5.2.2.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.3 Marking for shipment.- Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether Power Supply Unit RA-61-( ) is to be procured with or without Mounting FT-414 (see note below).
(c) Whether domestic or overseas shipment is required.
(d) Samples required.
(e) Marking and shipping of samples.
(f) Place of final inspection.
(g) Acceptable quality levels.
(h) Classification of defects.
(i) Facsimiles required.
(j) Running spare parts required.

Note: Mounting FT-414 is not part of Power Supply Unit RA-61-( ) and therefore must be procured as a separate item.

6.2 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter, identifying the particular design, for example: Power Supply Unit RA-61-W. As soon as possible after award of contract, the contractor should apply to the contracting officer for such information.

6.3 Contractual requirements.- It is recommended that the following be included in the contract:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:54Z
Check the source to verify that this is the current version before use.

6.3.1 Additional testing.- It should be understood that the additional inspection and non-destructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents. The Government inspector, therefore, may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection and testing, performed either at a Government Laboratory or the contractor's plant.

6.3.2 Waiver of preproduction samples.- When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that the specified samples have been submitted and approved on the current contract; production of the item under the new order will be concurrent with, or follow without interruption, production on the current order; and current production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent or renegotiation of the contract to reflect the saving in cost to the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:54Z
Check the source to verify that this is the current version before use.